Michael Chen

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Education

Purdue University — **Present – May 2027**
B.S. in Computer Science; Minor in Art & Design Studio — *West Lafayette, IN*

- **Cumulative GPA:** 3.94 | **Dean's List & Semester Honors:** 6/6 Semesters
- **Relevant Coursework:** Compilers, Data Mining & Machine Learning, Analysis of Algorithms, Artificial Intelligence, Numerical Methods, Fundamentals of Computer Graphics, Data Structures & Algorithms, Computer Architecture, Systems Programming, Programming In C, Discrete Math, Problem Solving & Object Oriented Programming

Experience

Itron Inc. — **May 2026 – Aug. 2026**
Full Stack Developer Intern — *San Jose, CA*

- Deployed functionality and designs for a new UI webpage built on an Angular framework with TypeScript, HTML5, and CSS that communicates via REST API calls with a back end involving C# and SQL.
- Leveraged AI tools to generate initial implementations for Jest unit tests, pinpoint flaws in the repository, and understand behavior of complex features.
- Resolved fatal Angular UI bugs within the live customer portal, enabling a more seamless user experience for key industrial energy management corporations.

SeeFood - Stanford Mike Snyder Lab Research (Abstract) — **May 2025 – Aug. 2025**
Software Engineer Intern — *Palo Alto, CA*

- Collaborated with multidisciplinary teams of research scientists and developers to engineer an accurate nutritional data logging platform.
- Co-authored a poster for research comparing GPT-4o-mini and SnapCalorie on their effectiveness of identifying ingredients and quantifying macronutrients of **35 total smartphone images of meals**.
- Developed a pipeline using Python and the OpenAI API to analyze **35 images** captured with Meta Ray-Ban Smartglasses, optimizing ingredient recognition consistency.

SLC - Stanford Mike Snyder Lab Research — **May 2024 – Aug. 2024**
Software Engineer Intern — *Palo Alto, CA*

- Collaborated with a team, including an assistant professor and graduate student, to analyze the effects of a Chinese herbal medicine on long COVID, based on a study involving wearable watch data from **19 patients over 8 weeks**.
- Processed **20GB** of time-series data on vital signs (heart rate, sleep, temperature), performing feature extraction in Python using tsfresh and catch22 to support machine learning applications.
- Generated PCA plots with scikit-learn and Matplotlib, filtering data by time and patient to visualize trends between experimental groups.

Projects

Personal Website (GitHub) — **Apr. 2026 – Present**

- Constructed a website with the Astro framework utilizing HTML5, JavaScript, TypeScript, CSS, Markdown, and MDX, which is deployed on GitHub Pages with GitHub Actions.
- Designed an elegant, dynamic, and interactive thematic with graphics powered by the shaders.com library that provides accommodation for lower end machines.

Gaming Bot - Discord Chatbot (GitHub) — **Aug. 2018 – Present**

- Independently developed and managed a Discord chatbot, deploying it to a server with **150 members**, enabling users to play games, listen to music, moderate servers, and use other tools such as a poll creator.
- Maintained long term functionality and modernized UI/UX by migrating legacy code to modern Discord API standards and regularly refactoring features.
- Published to a GitHub repository, coded in JavaScript within a Node.js environment, interacts with users via an intuitive Markdown UI, hosted externally on Oracle Cloud, and utilizing a database with JSON files, previously hosted on Heroku with a PostgreSQL database.

Additional

Programming Languages: Python, C, C++, Java, JavaScript, TypeScript, LaTeX, Markdown, Matlab, HTML5, CSS
Frameworks/Libraries: tsfresh, scikit-learn, Matplotlib, catch22, Node.js, Pandas, NumPy, OpenGL, Astro, Angular, REST APIs
Infrastructure: Linux, Oracle Cloud, PostgreSQL, Heroku, Git, GitHub
Awards: USA Coding Olympiad Gold Status, Homestead High School Valedictorian